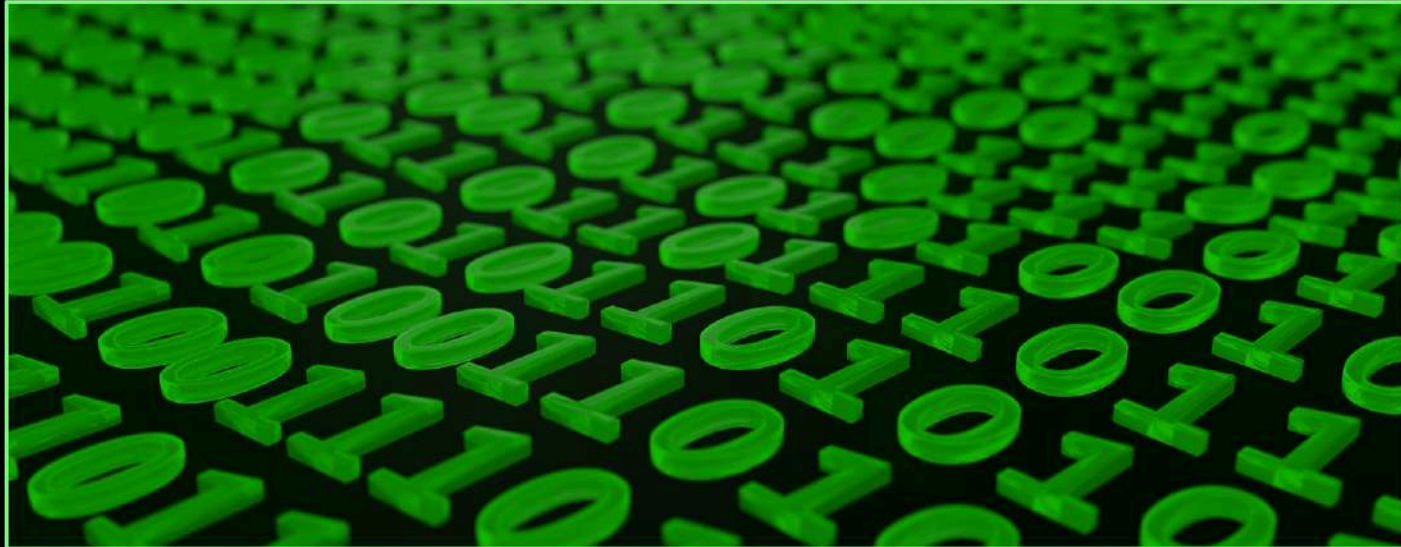


CHAPTERS 10 & 11 OF BECOMING A DATA HEAD



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9/10/2026

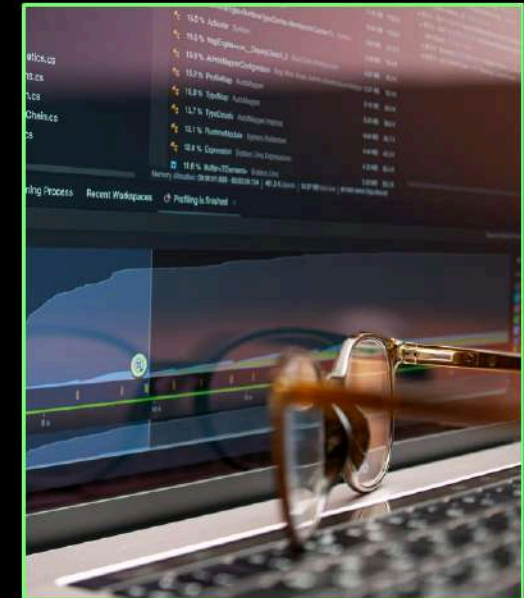
THE STORY OF THESE TWO CHAPTERS

Classification predicts categories. Text analytics turns language into usable data.

Chapter 10
Classification

Probabilities
& decisions

Chapter 11
Text as numbers



FROM REGRESSION TO CLASSIFICATION

REGRESSION

Predict a number

House -> \$350,000

Sales -> \$12,400

CLASSIFICATION

Predict a category

Email -> Spam / Not Spam

Applicant -> Interview / No

CLASSIFICATION IS NOT CLUSTERING

CLUSTERING

No labels beforehand

Discover groups

You name the groups later

CLASSIFICATION

Labels already exist

Learn from examples

Predict known categories

BINARY VS. MULTICLASS

BINARY

Two possible classes

Default / No Default

Disease / No Disease

MULTICLASS

More than two classes

Honda / Toyota / Ford

News: Sports / Politics / Local

Positive class = the class we are looking for

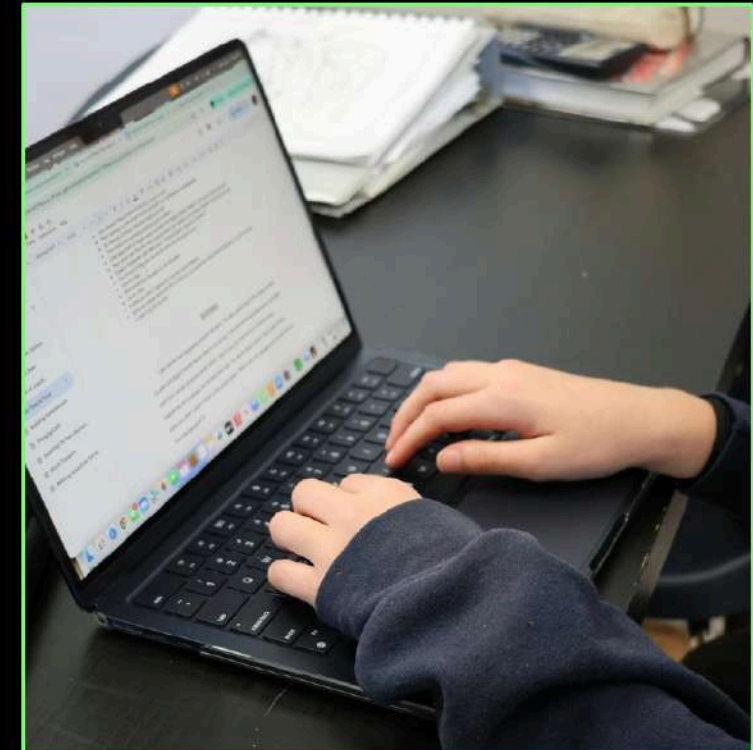
THE INTERNSHIP SELECTION EXAMPLE

Historical data gives us features and a target label.

Features: GPA, year, major, extracurriculars

Target: Interview offer? Yes / No

Goal: predict whether a new applicant should be offered an interview



WHY LINEAR REGRESSION IS NOT ENOUGH

A probability cannot be less than 0 or greater than 1.

Encode: Yes = 1, No = 0

Linear regression might output -0.1 or 1.3

Those are meaningless as probabilities

$$0 \leq P \leq 1$$

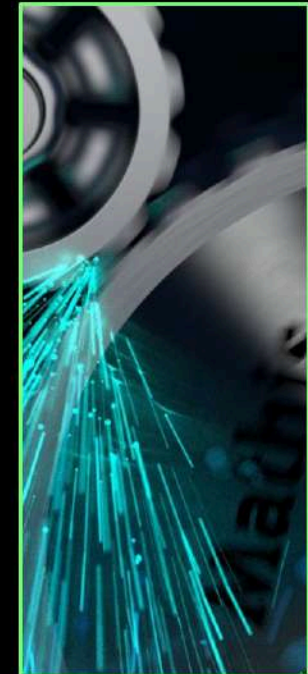
LOGISTIC REGRESSION GIVES PROBABILITY

The output is not the final label - it is a probability.

Applicant
features

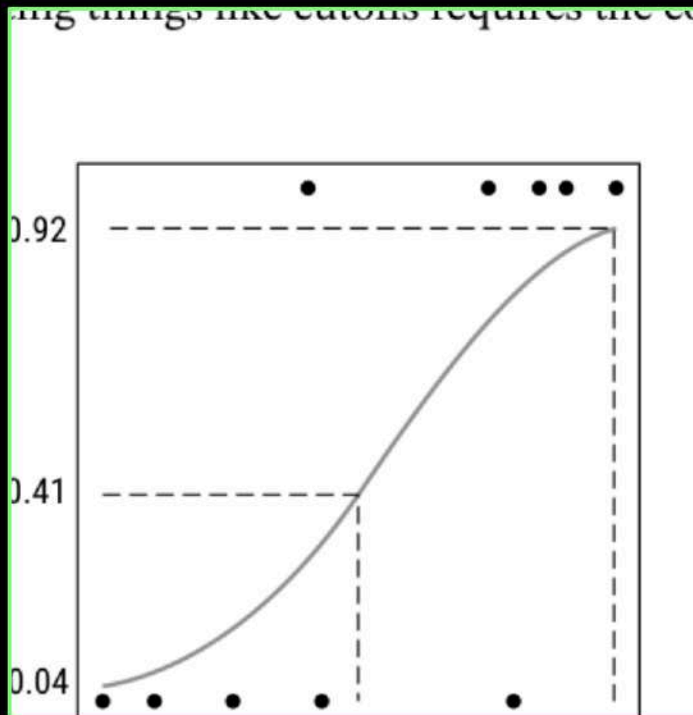
Logistic
model

$P(\text{Interview})$
 $= 0.82$



THE S-CURVE INTUITION

The logistic function squishes outputs between 0 and 1.



Linear regression: straight line, unbounded

Logistic regression: S-shaped probability curve

A one-unit change does not always have the same effect

GPA TO PROBABILITY

2.0 GPA -> 4%

3.0 GPA -> 41%

4.0 GPA -> 92%

Same one-point change, different probability jump.

That is the shape of logistic regression.

PROBABILITY IS NOT YET A DECISION

The model says how likely. Humans decide what to do.

Applicant A -> 74%

Applicant B -> 46%

Applicant C -> 91%

Still need a decision rule



DECISION THRESHOLD

Example cutoff: 0.50

If $P \geq 0.50 \rightarrow \text{YES}$

If $P < 0.50 \rightarrow \text{NO}$

***0.50 is a default,
not a law.***

CHANGING THRESHOLD CHANGES MISTAKES

LOWER THRESHOLD

More YES predictions

More false positives

Fewer false negatives

HIGHER THRESHOLD

Fewer YES predictions

Fewer false positives

More false negatives

THRESHOLD IS A DOMAIN DECISION

Should these use the same cutoff?

Marketing email campaign

Expensive medical treatment

Cancer screening



Different errors have different costs

DECISION TREES

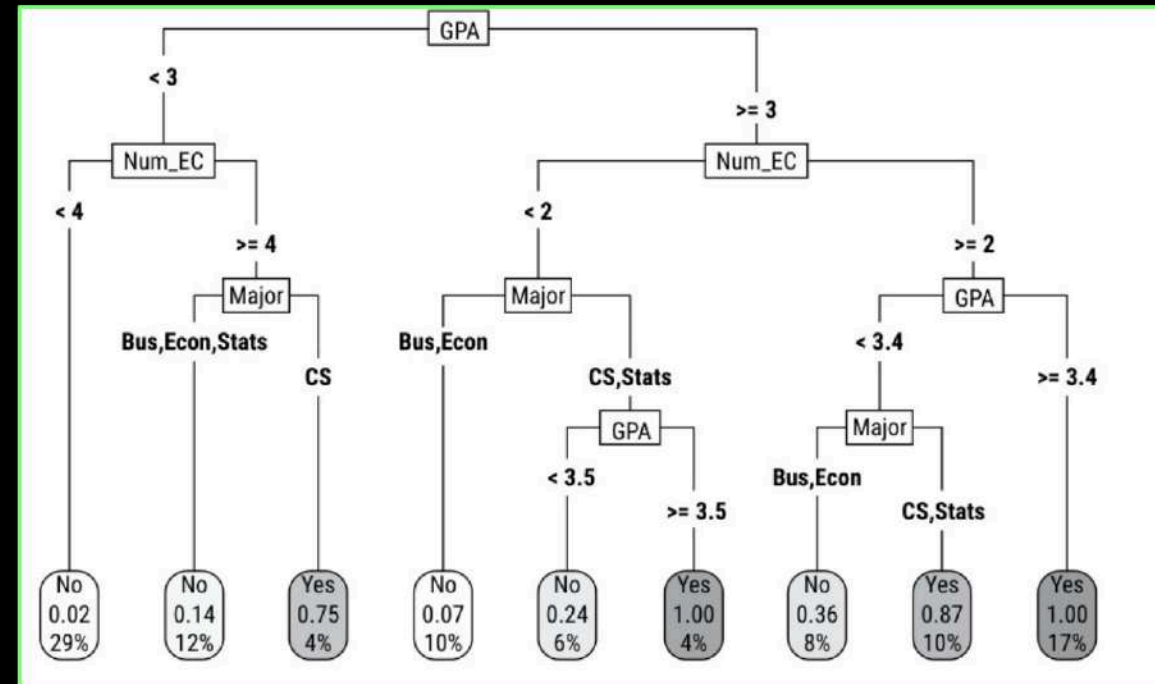
A decision tree creates rules from the data.

GPA ≥ 3.0 ?

Extracurriculars ≥ 3 ?

Major = CS?

Interview / No Interview



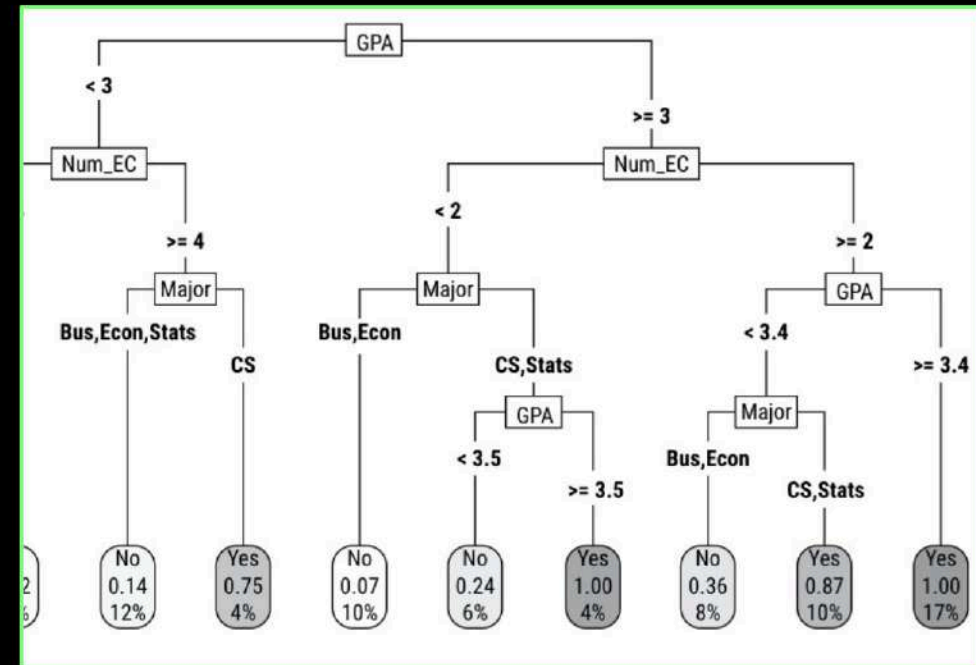
HOW A TREE MAKES A PREDICTION

Start at the root and follow the branches.

Applicant: GPA 3.6, Year 2, Major CS, EC 1

Follow the path through the tree

Leaf prediction -> YES



TREES CAN DISCOVER INTERACTIONS

A feature does not always act alone.

Low extracurriculars might be offset by:

High GPA

CS or Statistics major

A specific year in school

The rule depends on the path

BUT TREES OVERFIT EASILY

Keep splitting...
and splitting...
until every applicant has their own leaf.

***Perfect training fit
can be a warning.***

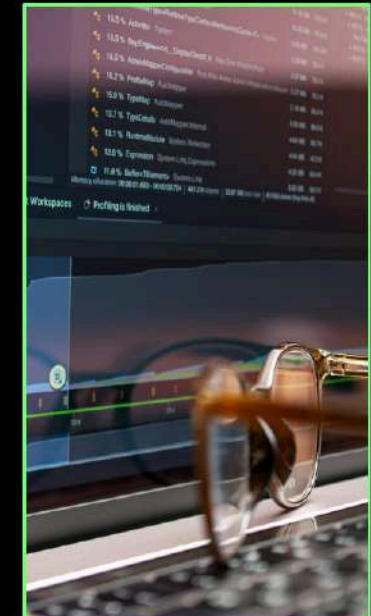
ENSEMBLE METHODS

Instead of trusting one model, combine many models.

One tree
unstable

Many trees
combined

More robust
predictions



RANDOM FOREST: MANY OPINIONS

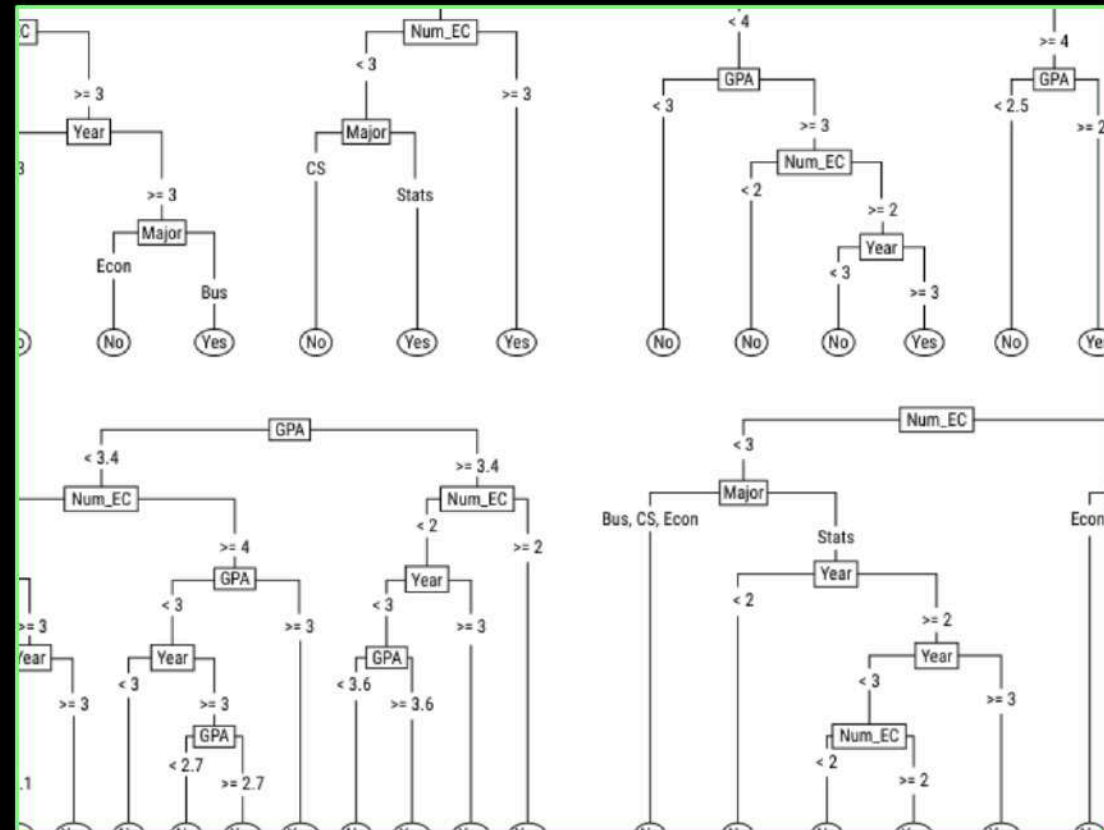
Tree 1 -> YES

Tree 2 -> NO

Tree 3 -> YES

Tree 4 -> YES

Majority vote -> YES



WHY IS IT RANDOM?

Each tree sees a slightly different world.

RANDOM ROWS

Different applicants

Different samples

Less dependence on one dataset

RANDOM FEATURES

Different variables

Different split candidates

Less redundant trees

GRADIENT BOOSTING

Many corrections accumulated.

Tree 1

Find errors

Tree 2
focuses on errors

Tree 3
improves again

Unlike random forests, boosted trees are built sequentially.

ACCURACY VS. INTERPRETABILITY

EASIER TO EXPLAIN

Logistic regression
Single decision tree
Clearer rules

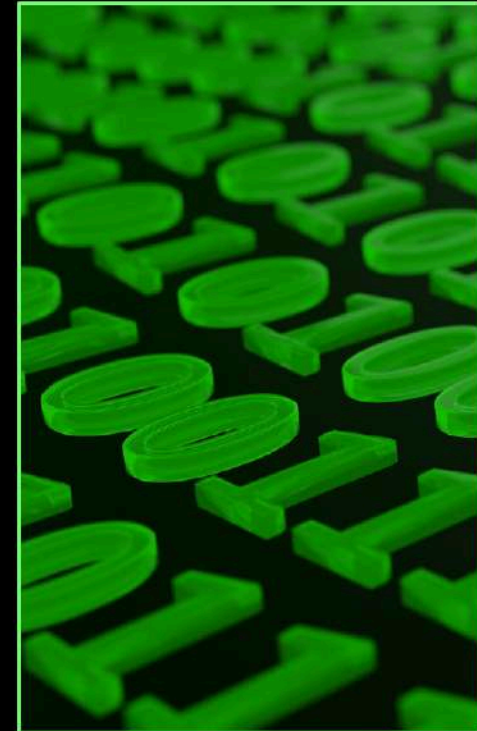
OFTEN STRONGER

Random forests
Boosted trees
Harder to explain

Predictive power often trades off with interpretability

CLASSIFICATION PITFALLS

- Misapplying the problem
- Data leakage
- Not splitting train/test data
- Choosing a bad threshold
- Misunderstanding accuracy



ACCURACY CAN LIE

A useless model can look great with imbalanced data.

99.5%

of users do NOT click an ad.

Predict "No click" for everyone.

Accuracy = 99.5%

Useful? No.

ALWAYS COMPARE AGAINST A BASELINE

Intern data: 60% = No Offer

Dumb baseline: predict No for everyone

Baseline accuracy = 60%

Model accuracy = 60%

***Model added
nothing.***

CONFUSION MATRIX

	Actual YES	Actual NO
Predicted YES	True Positive	False Positive
Predicted NO	False Negative	True Negative

It shows how the classifier is right or wrong.

Different mistakes can have different costs.

FALSE POSITIVE VS. FALSE NEGATIVE

FALSE POSITIVE

Predicted YES

Reality = NO

Example: flag good email as spam

FALSE NEGATIVE

Predicted NO

Reality = YES

Example: miss a sick patient

RECALL AND SPECIFICITY

RECALL / SENSITIVITY

Of the actual positives:

How many did we catch?

High recall = fewer missed positives

SPECIFICITY

Of the actual negatives:

How many did we reject correctly?

High specificity = fewer false alarms

CHAPTER 10 TAKEAWAY

Classification is about probabilities, decisions, and mistakes.

A good Data Head asks: Which threshold? Which metric? Which error matters more?



CHAPTER 11: TEXT ANALYTICS

Now the input is not a clean table - it is language.

Emails

Reviews

Articles

Comments

Social media posts



STRUCTURED VS. UNSTRUCTURED DATA

STRUCTURED

Rows and columns

Age, income, purchases

Ready for statistical tools

UNSTRUCTURED

Sentences and paragraphs

Emails, reviews, notes

Needs transformation first

COMPUTERS NEED NUMBERS

Humans read meaning.

Computers need numbers.

Sarcasm, mood, context, and ambiguity do not automatically survive the conversion.



TEXT ANALYTICS PIPELINE



Every step is a choice. Every choice can preserve or destroy information.

BAG OF WORDS

The simplest way to turn text into numbers.

Sentence: "Data science is fun"

data -> 1

science -> 1

is -> 1

fun -> 1

Word order is ignored

DOCUMENT-TERM MATRIX

Document	good	bad	price	product
Review 1	2	0	1	1
Review 2	0	3	0	1
Review 3	1	0	2	1

Rows = documents

Columns = tokens

Values = counts

TEXT CREATES SPARSE HIGH-DIMENSIONAL DATA

***100,000
unique words***

becomes 100,000 features.

Most documents use only a tiny subset.

The matrix becomes wide and full of zeros.

TEXT PREPROCESSING

Lowercase

**Remove
punctuation**

**Remove
stop words**

**Stem or
lemmatize**

**Count
tokens**

Cleaning makes the data smaller and easier to analyze. But it is not neutral.

THE HOTDOG PROBLEM

Jordan loves hotdogs but hates hamburgers.

Jordan hates hotdogs but loves hamburgers.



***Same words.
Opposite meaning.***

N-GRAMS ADD SOME CONTEXT BACK

UNIGRAMS

loves

hotdogs

hates

BIGRAMS

loves hotdogs

hates hotdogs

meaning becomes clearer

More context = wider matrix

WORD EMBEDDINGS

Words become vectors of numbers.

cow = (1.0, 0.1, 1.0)

pig = (1.0, 0.1, 0.0)

pork = (0.1, 1.0, 0.0)

Similar usage -> similar vectors

EMBEDDINGS CAPTURE RELATIONSHIPS

Beef - Cow + Pig $\approx$ Pork

The math works because the vectors encode patterns of word usage.

CONTEXT STILL MATTERS

GENERAL LANGUAGE

Goldfish -> animal / pet

Tide -> ocean

GROCERY DATA

Goldfish -> crackers / snack

Tide -> detergent

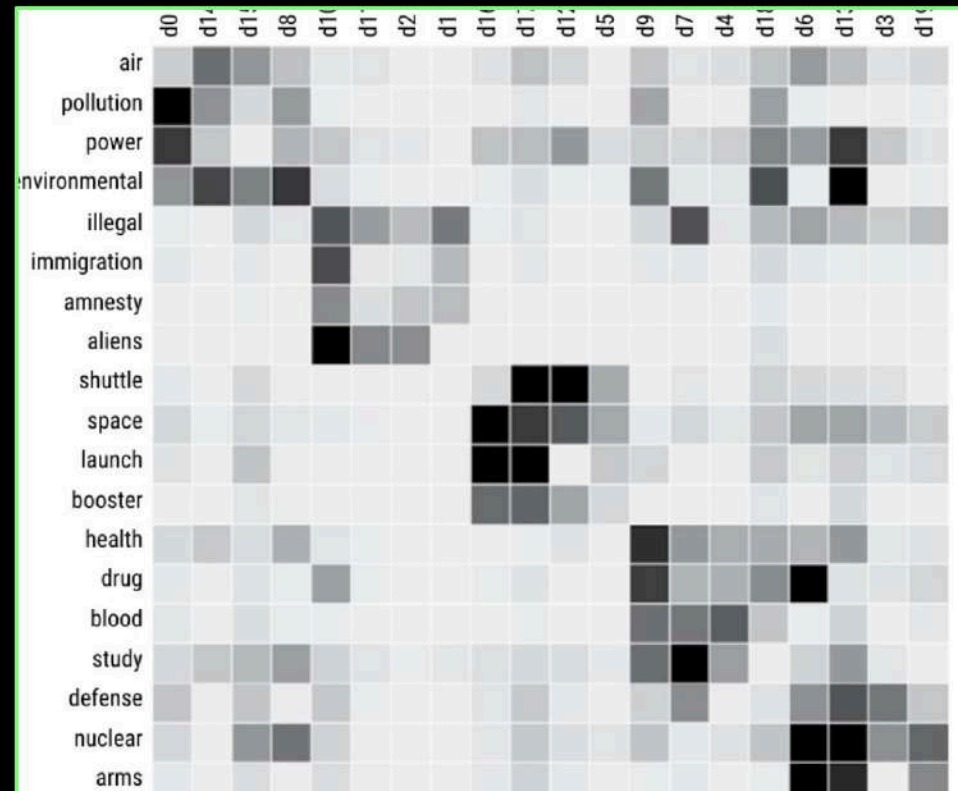
TOPIC MODELING

A document can belong to more than one topic.

Article A:

60% Defense

40% Space



TOPIC MODELING REQUIRES JUDGMENT

No guarantee of clean topics

Corpus diversity matters

Filtering too much can remove the structure you hoped to discover

Unsupervised learning still needs human judgment

TEXT CLASSIFICATION

Now Chapter 10 and Chapter 11 connect.



When you mark an email as spam, you are helping create labels for future models.

SENTIMENT ANALYSIS



It is text classification.

But a model trained on movie reviews may fail on specialized business language.

Training context defines what the model has learned.

BIG TECH LOOKS LIKE MAGIC

Set expectations for text analytics.

Huge datasets

Huge labeled datasets

Powerful compute

Specialized research teams

Money



WORD CLOUDS: PRETTY, BUT USEFUL?



Word size shows frequency.

But area is hard to compare.

A sorted bar chart is usually clearer.

CONNECTION BETWEEN CHAPTERS 10 AND 11

CHAPTER 10

Structured features

Classification

Category prediction

CHAPTER 11

Raw text

Turn it into numbers

Classification / topics



FINAL TAKEAWAY

***Models calculate.
Humans define what counts
as a good decision.***

Thresholds, mistakes, text representations, and business context are human responsibilities.



THANK YOU

QUESTIONS & DISCUSSION



CHAPTERS 10 & 11

